

Zihao Jing

Western University, Canada

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I am a Master's student in Computer Science at Western University, Canada. My work focuses on LLM post-training, multimodal alignment, and model adaptation. I have first-author publications at NeurIPS, ICLR, and ICML, along with prior industry experience in LLM training and text embedding models at SenseTime.

EDUCATION

Western University, M.Sc. in Computer Science, Ontario, Canada	Sep. 2024–Jun. 2026
Beihang University, B.Eng. in Software Engineering, Beijing, China	Sep. 2019–Jun. 2024

SELECTED PUBLICATIONS

Top-Tier Conferences

- [1] **Zihao Jing**, Qiuhan Zeng, Ruiyi Fang, Yan Yi Li, Yan Sun, Boyu Wang, Pingzhao Hu. *Scaling-Aware Adapter for Structure-Grounded LLM Reasoning*. **ICML 2026** [Code] [Paper]
- Designed an entropy-guided adapter that selects instruction-relevant structural regions for variable-size inputs, addressing uniform treatment of positions in structure-grounded LLM reasoning. Achieved top-1 performance on 17 out of 18 structured reasoning benchmarks.
- [2] **Zihao Jing**, Qiuhan Zeng, Ruiyi Fang, Yan Sun, Boyu Wang, Pingzhao Hu. *Entropy-Guided Dynamic Tokens for Graph-LLM Alignment in Molecular Understanding*. In **ICLR 2026**. [Code] [Paper] [Poster] [Video][Slices]
- Built a dynamic connector that scales representation slots based on input complexity, reducing structural information loss from fixed-size connectors. Ranked #1 on 20 out of 21 interdisciplinary understanding tasks and trained 3.5× faster under equivalent settings.
- [3] **Zihao Jing**, Yan Sun, Yan Yi Li, Sugitha Janarthanan, Alana Deng, Pingzhao Hu. *Structure-Aware Fusion with Progressive Injection for Multimodal Molecular Representation Learning*. In **NeurIPS 2025**. [Code] [Paper] [Video&Poster]
- Built a multimodal encoder that stabilizes noisy structured inputs into reliable representations, improving robustness in multimodal embedding with geometric features. Ranked #1 on 22/29 interdisciplinary downstream benchmarks with up to 27% better accuracy.

Selected Additional Works

- [4] Junqin Huang, Zhongjie Hu, **Zihao Jing**, Mengya Gao, Yichao Wu. *Piccolo2: General Text Embedding with Multi-Task Hybrid Loss Training*. **SenseTime Technical Report, 2024** [SenseNova] [Code] [Report]
- Built training/evaluation pipelines and contributed to iterative optimization of a general-purpose embedding model; achieved top-1 on C-MTEB in May 2024.

Additional Co-authored Publications

Co-authored 3 additional papers at **ICML 2026** and **ICLR 2026** (×2).

RESEARCH INTERNSHIPS

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| SenseTime | <i>LLM Research Intern</i> | 2023.09–2024.06 |
| <ul style="list-style-type: none">Developed Piccolo-GPT, supporting text embedding and generation within a single LLM architecture.Built training/evaluation pipelines for text embedding models (Piccolo2), achieving top-1 on C-MTEB (May 2024).Fine-tuned a ~100B LLM for vertical livestream marketing script generation at Sina Weibo. | | |
| Jina AI (acquired by Elastic) | <i>AI Research Intern</i> | 2023.04–2023.09 |
| <ul style="list-style-type: none">Implemented LLM-based denoising and sentiment analysis pipeline for Budweiser; reduced cost >13%.Led the evaluation and deployment of super-resolution models for commercialization.Integrated model inference pipelines into open-source LLM tooling in the LlamaIndex ecosystem. | | |

RESEARCH GRANTS

Digital Research Alliance of Canada, RRG Competition 2026 — Secured 5×A100-80GB GPU-years on Canada's national supercomputing cluster (~US\$80K commercial value).

TECHNICAL SKILLS

- Research:** LLM post-training, pre-training/fine-tuning, and multimodal alignment.
- ML Engineering:** PyTorch, Transformers, DeepSpeed, DDP/FSDP, profiling, and monitoring.
- Systems and Programming:** Linux, Slurm, Docker, Python, C/C++, Java, and SQL.

SELECTED AWARDS AND HONORS

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| • Third Prize , 13th National Undergraduate Mathematics Competition (National-level) | 2021 |
| • Third Prize , 32nd Beijing Undergraduate Mathematics Competition (Province-level) | 2021 |
| • Honorable Mention (H Prize) , Mathematical Contest In Modeling | 2020 |